Four dimensional Yang–Mills research dossier

Research handoff

17 October 2026

Contents

1 Purpose, conclusion and evidence rules **3**
- 1.1 What a proof claim in this bundle means . . . 3
- 1.2 The archive's order of authority . . . 3

2 The target and the hierarchy of conclusions **4**
- 2.1 Distinct parameters and uniformities . . . 4

3 Repository baseline, mathematical provenance, and additions **4**
- 3.1 Frozen evidence and reproducibility . . . 4
- 3.2 What the README and zeroth manuscript claim . . . 5
- 3.3 Proof ledger and unresolved version crosswalk . . . 7
- 3.4 Navigation through the original mathematics . . . 8
- 3.5 Published results and transfer obligations . . . 9
- 3.6 Our additions: objects, results and boundaries . . . 9
- 3.7 Corrections and superseded routes . . . 12
- 3.8 Remaining existence interfaces and audit priorities . . . 14
- 3.9 Companion files and citation conventions . . . 14

4 The local construction added in this project **15**
- 4.1 Flat and nonlinear classical analysis . . . 15
- 4.2 The actual local quantum integral . . . 15
- 4.3 Ward extraction and finite local induction . . . 16
- 4.4 The actual logarithmic drift . . . 16
- 4.5 Finite weak histories and the recursion actually tuned . . . 17

5 The full-measure additions **17**
- 5.1 Exact positive history operators . . . 17
- 5.2 Actual Wilson rarity and conditional normalization . . . 18
- 5.3 The final normalized replacement . . . 18

6 The remaining theorem program **19**
- 6.1 H2 Complete-density transfer for the chosen map . . . 19
- 6.2 H3 Observable and source estimates . . . 19
- 6.3 H4 A common continuum hierarchy . . . 19
- 6.4 H5 Positivity, rotations and reconstruction . . . 20
- 6.5 H6 Nontrivial Yang–Mills identification . . . 20
- 6.6 H7 Physical mass gap and group scope . . . 20

7 How a successor can use the bundle **20**
7.1 A recommended first pass 20
7.2 A concrete next research deliverable 21
7.3 Review and reproducibility 21

1 Purpose, conclusion and evidence rules

This dossier transfers a sustained attempt to complete a rigorous existence argument for a nontrivial four-dimensional Yang–Mills quantum theory. The project starts from the repository *Yang–Mills-LLM-experiment*, hosted at `ethanwsussman/Yang-Mills-LLM-experiment` with a README note signed LA, and from the Bałaban renormalization literature. The detailed baseline assessment below distinguishes the repository's stated ambition from the precise scope of its displayed results.

At handoff there is a substantial local ultraviolet construction in the working proof chain, together with new exact statements about full finite Wilson measures. There is not yet a completed continuum existence proof. A complete-density multiscale theorem, observable estimates, and continuum identification still require mathematical arguments.

The final research step, Notebook 39, completes a restricted normalized replacement operation on arbitrary collections of separated fixed boxes in the actual bare Wilson theory. Its denominator is controlled, its exterior marginal is preserved, and its effect on local bounded observables does not grow with the number of distant boxes. This is a natural stopping point for the present round: one finite-measure operation is complete, and its remaining distance from a multiscale construction is explicit.

1.1 What a proof claim in this bundle means

The documents distinguish five kinds of statement.

1. A *derived working result* has a written argument in the included notebooks, with its assumptions and scope stated. Many recent modules have separate internal mathematical audits. This is not independent professional verification of all earlier dependencies.

2. A *conditional reduction* proves an implication from explicit model bounds. Its hypotheses are not counted as established merely because the reduction is proved.

3. A *published input* is tied to an actual primary statement. Its use requires matching its domains, normalizations and uniformities.

4. An *open interface* is a step of this proposed route for which no complete argument is supplied here. A different route to existence might replace it.

5. A *superseded argument* is retained for provenance, with its correction or replacement identified. Its presence in the research archive is not an endorsement.

The terms our addition and derived here identify contributions to this project record. They are not claims of novelty relative to the whole literature. The compact-group Laplace, tree-gauge, coupling and chessboard mechanisms have standard antecedents. Their precise assembly and use in the present program are what this dossier records.

1.2 The archive's order of authority

Read the final handoff documents for the current map, then the numbered notebooks for statements and proofs. Read each notebook's dependencies before treating its theorem as unconditional. The research archive contains fragments, calculation notes, audits and historical checkpoints. Later corrections override earlier speculation and earlier next-task paragraphs.

The catalogue supplies an entry for every numbered notebook: its location, central results, assumptions, dependencies and limitations. The manifest identifies exact files by SHA-256. The repository is pinned separately; a future remote update is not silently substituted for the examined commit.

2 The target and the hierarchy of conclusions

The working regulator is a finite four-dimensional periodic Wilson lattice gauge theory, principally for $SU(2)$ in the inherited construction. At positive coupling its compact group integral exists automatically. The hard question is a nontrivial continuum theory at vanishing lattice spacing, with the quantum-field properties needed to identify it as Yang–Mills.

Three conclusions must be distinguished. A continuum limit of separated scalar action-density correlations is a useful observable sector. Reconstruction from a sufficient Euclidean hierarchy could give a local relativistic quantum observable theory. Identifying it as nontrivial Yang–Mills requires an adequate gauge-invariant algebra and its relation to the lattice theory and short-distance normalization. A physical mass gap is a further spectral assertion. A result for $SU(2)$ does not by itself cover every compact simple gauge group.

Subsequential existence is a legitimate target. A unique cutoff limit or preferred contact-term convention is not imposed solely for convenience. What is indispensable is one consistent limiting hierarchy with the requisite positivity, covariance, regularity and physical interpretation.

2.1 Distinct parameters and uniformities

In the main local chain, M denotes preceding homogeneous classical depth; ℓ is the fixed primitive ratio; $R = \ell^s$ is a fixed composite quantum ratio; J is a fixed auxiliary grouping; N is a finite period; b is a small analytic radius; g is the running coupling; and K is a large endpoint ratio in determinant comparisons. In physical lattice statements η denotes mesh size. Individual theorem statements control any changes of notation.

Uniformity in M, N on a fixed small ball is not uniformity on arbitrary exterior fields. Constants depending on fixed R, J are not uniform as either grows. A fixed-box bound uniform in surrounding volume is not a bound on a physical box containing more and more lattice sites. An infinite structural kernel is not automatically the actual finite winding kernel. Required old-depth shifts, endpoint ratios and source derivatives must belong to one joint structural limit.

3 Repository baseline, mathematical provenance, and additions

Evidence date: 17 October 2026. This report distinguishes what the deposited repository claims; what its displayed arguments take as inputs; what published literature supplies in its own setting; and what this research series adds. It is a reading and dependency report, not certification of either the repository or our notebooks. Neither manuscript length, a theorem environment, successful compilation, nor repeated internal review establishes a proof.

The repository claims an apparent existence argument on $\mathbb{R}^4$. Describing the baseline as only finite-volume work, only a programme, or no claimed existence argument would be inaccurate. Identifying that claim with an externally verified Yang–Mills construction with a mass gap would also be inaccurate. The shorter manuscript expressly states a narrower observable theorem, records conditional deductions and excludes stronger conclusions. Our additions address selected interfaces and develop a separately specified local construction. They do not retrospectively verify every argument in the deposit.

3.1 Frozen evidence and reproducibility

The source is `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, frozen at `e392ef45013a581664bab76b90867170b30fe10e`. The commit is dated 16 October 2026, 02:33:55 −0400, has message *Update README.md*, and records the Git author as alpoge. This identifies a version, not authorship of every generated or human-written argument. The raw master has an empty

author field. We do not infer a formal author list from the hosting account or Git metadata. The checkout was clean when checked for this report and was not edited. Later remote changes are outside the evidence boundary.

Raw line numbers are counted from one in that commit. PDF locators are printed pages of `ym4_manuscript_zeroth_draft.pdf`, dated 6 October 2026. In the local extraction its page markers agree with printed pages. Raw labels are preferable to generated theorem numbers: the raw master does not presently assemble unchanged. Its lines 38–40 input `theorem_0_complete_account_not_done_for_ethan_v2_part1`, `theorem_0_complete_account_not_done_for_ethan_v2_part2`, and `theorem_0_complete_account_not_done_for_ethan_v2_part3`, whereas the actual files are `theorem_0_complete_account_raw_I.tex`, `theorem_0_complete_account_raw_II.tex`, and `theorem_0_complete_account_raw_III.tex`. This is a reproducibility naming issue, not a mathematical counterexample. The handoff preserves it rather than silently repairing third-party source.

The three raw parts have respectively 166,711, 145,624 and 111,583 lines: 423,918 in total. These counts measure material to audit, not proved mathematics. The companion `handoff_repository_source_map.json` records commit, SHA-256 hashes, byte counts, exact locators and selected links. The handoff uses a manifest and retrieval instructions rather than redistributing third-party source PDFs or articles.

Below, I, II and III mean the three raw files; Draft means the zeroth PDF; Our Notebook always means a document of this research series, not a repository document.

3.2 What the README and zeroth manuscript claim

README, lines 1–6. The opening describes ongoing model-assisted QED3, YM3 and YM4 work, with exposition and formalization in progress. The note at lines 5–6, signed LA, explicitly reports an apparent YM4 existence argument in $\mathbb{R}^4$, emphasizes its length and lack of readability, and explains the simpler presentations as a way of exposing ideas, including a reflection-positivity route to rotations. This is a qualified public claim, not a theorem statement or a mass-gap claim.

The pinned tree also contains `QED3_slop/YM_Rotation_Invariance_simplified.pdf`, `QED3_slop/qed3_manuscript.pdf`, and `QED3_slop/qed3.pdf`. Their presence and the README already place the reflection-based rotation idea in the baseline. This report does not independently certify those lower-dimensional manuscripts or treat our pure-Wilson interface work as the origin of that idea.

Draft, abstract and Sections 1.1–1.4, pp.1–23. The object is the Wilson lattice measure for $SU(2)$, on compatible periodic tori, and correlations of one local gauge-invariant observable: smeared action density. Supports of the test functions are pairwise disjoint. The density has canonical dimension four; its leading separated two-point kernel has singularity $|x-y|^{-8}$. There is no unmentioned prescription extending it across coincidence.

In the draft's convention,

$$A(U) = \sum_p (1 - \operatorname{Re}\operatorname{tr} U(\partial p)), \qquad d\mu(U) \propto e^{-g_0^{-2} A(U)}\, dU, \qquad \operatorname{tr} 1 = 1.$$

The observable is the corresponding test-weighted plaquette sum. Complex sources change action weights; connected correlations are derivatives of the logarithm of the sourced partition function, and centred moments are assembled from cumulants. This convention matters when comparing beta coefficients and two-point amplitudes. See Draft pp.7–9 and I, Chapter 1, beginning at line 401.

Theorem 0: scope of the clauses. This is a navigation map, not a substitute for the parameter hypotheses in the displayed theorem.

(0), running coupling. The draft asserts an actual recursively generated trajectory stopped at a first-passage scale, with two-sided cumulative inverse-coupling bounds and, for $SU(2)$, a cumulative one-loop law. It does not require a positive decrement at every step. I:2533–2554 explicitly distinguishes cumulative drift from a single-step sign.

(i), ultraviolet stability. The application of Bałaban's complete-density theorem uses the weak-coupling containment furnished by (0). Stability concerns effective densities and their prescribed representation, not by itself a continuum observable limit.

(iii), sourced correlations. Connected action-density bounds are asserted uniformly in cutoff; a separate per-ball estimate supplies the needed volume independence. This source theorem and its running shift are substantive inputs, not consequences of a vacuum partition-function estimate alone.

(ii), limits and reconstruction. The claim includes torus and $\mathbb{R}^4$ limits with the stated fixed-order regularity, translations, hypercubic symmetry, permutation symmetry and reflection positivity. It reconstructs a Hilbert space, vacuum and nonnegative time-translation generator in the stated sense. The $\mathbb{R}^4$ limit is subsequential. Fixed-torus convergence without subsequences and identification of first-passage limits use (v), inheriting its dependencies.

(iv), a two-point witness. At small separated distances the connected two-point function is comparable to the specified positive free curvature kernel times the fourth power of a running coupling, with controlled relative error. A one-torus witness is transferred to other volumes. Its consequences include a nonzero centred reflected vector and logarithmic short-distance behavior.

(v), a torus family. Under comparison inputs, a unique analytic one-parameter family is asserted at fixed torus size. Two-cutoff and first-passage/tuned identifications are stronger than subsequential compactness. They are not infinite-volume uniqueness or clustering.

(vi), rotations. Full $O(4)$ invariance is conditional on a displayed far-sphere angular-momentum-flux hypothesis, which the draft does not establish. Short-distance approximate rotation statements and the README's later reflection argument are not interchangeable assertions.

Exact reading locations are Draft Theorem 0, pp.12–23; Sections 2, 3, 4, 5–6, 7 and 8 for (0)–(i), (iii), (ii), (iv), (v) and (vi), respectively. The raw main statement is I:2512, `3.1:thm-main`; its dependency assembly is III:110784–110860, `ch:11`.

The meaning of non-triviality is deliberately limited. Draft p.5 and I:2475–2510, `3.1:def-non-trivial`, distinguish the positive, logarithmically running two-point law from stronger interaction assertions. The raw definition expressly permits a generalized-free realization of an admissible two-point function and observes that a composite of a free field need not be Gaussian. Our counterexamples and spectral-witness discussion must not be presented as discovering a limitation that the baseline denied. They clarify what a stronger Yang–Mills identification would require.

Explicit exclusions. Draft pp.5–6 and 150–151 do not claim a mass gap, clustering, uniqueness of the infinite-volume limit or vacuum, coincidence extensions, Wightman fields with the required growth in the number of arguments, Wilson-loop or block-observable laws, unconditional full rotations, or the complete result for $SU(N)$, $N \geq 3$. Its small-torus spectral discussion and finite-size criterion in the outlook are not a mass gap on $\mathbb{R}^4$. The title's construction and non-triviality language has these express limitations.

3.3 Proof ledger and unresolved version crosswalk

The hypotheses (H1)–(H7) in Draft Section 1.3 organize geometry, thresholds and quantifier order. They are not all the dynamical inputs. Section 9, pp.141–151, is the additional ledger: published citations, statements used as Bałaban asserts them, companion proofs, deferred proofs and unfinished supporting statements are separated.

What the shorter draft says is written. Draft pp.6 and 143 says that no own input below $SU(2)$ clauses (0)–(iv) is deferred, except portions of (ii) using (v). It claims companion proofs for the source theorem, witness decomposition, Gaussian extraction and one-cell large-field estimates, among others. Section 9.1 nevertheless records fifteen imported statements/readings without a printed proof covering their asserted use. A written companion proof and independent verification of every imported generality are different propositions.

What particular raw sites say. III:110862–110870 explicitly says no clause is proved outright. III:111167–111179 organizes outstanding proofs into five categories. This concerns implications from named inputs; it does not mean no part contains a proof. Two important exact sites are:

1. I:3901–3945, `4.2:lem-one-loop`, states the finite Gaussian one-loop law by citation to CMP109/CMP116 and begins an outline: $\left|\sum_{l=1}^{k} \beta_l^0 - k c_0\right| \le c_2$. I:3895–3898 gives $c_0 = 11N^2(12\pi^2)^{-1}\log L$ in the deposited normalized-trace convention.

2. I:4031–4088, `4.2:lem-unmarked`, marks the CMP116 remainder transfer to the CMP119 cutoff and fluctuation radius as an incomplete step. I:4136–4198, `4.3:thm-small-field-step`, displays the stronger family-form/cutoff reading. I:56419–56440, `4.19:prop-clause-zero`, lists the relevant assumptions; lines 56428–56430 call both one-loop and unmarked statements incomplete, although the current one-loop lemma instead uses the citation/outline description above.

The shorter draft's completion claim and these raw markers are not reconciled by a versioned proof crosswalk in the pinned files. We do not infer which supersedes which, or that a proof cannot exist outside the deposit. A reviewer should request the authoritative companion revision and exact replacements for remaining markers. Conversely, counting every outstanding string as an independent necessary gap would be wrong: some belong to optional routes, stronger conclusions or statements explicitly not used by the active route.

The fifteen imported readings. Draft pp.142–143 labels them Pr1–Pr15. In order, their subjects are:

1. large-field normalization/Wilson-term bounds, CMP122-II (1.11), (1.23)–(1.48);

2. analyticity of old terms beside a new region, p.375;

3. the resummed boundary estimate, p.376;

4. locality and hull bookkeeping, pp.379, 390;

5. an estimate whose details are omitted, p.360;

6. interrupted integrations, CMP122-I p.191;

7. prepared-data admissibility, CMP122-I pp.177–179;

8. regularity on consecutive small-field differences, CMP119 p.275;
9. transfer to a multiscale cover, CMP119 pp.276–277;
10. complex-domain inclusion, CMP109 (1.17);
11. a second regularity inequality, CMP109 p.279;
12. companion estimates asserted by analogy in the propagator/averaging papers and CMP109 (3.26);
13. lowest-tower estimates, (3.35), (3.37) of the draft's reference [5];
14. a background-field estimate derived from an uncomputed display, reference [6], (189)–(190);
15. regularity of the unintegrated large-field core, used for uniqueness.

This is the draft's evaluation of its imports, not our verdict that the originals are false or that proofs cannot exist elsewhere. Domains and uniformities must be compared in each application.

The uniqueness ledger. Draft pp.148–149 identifies three written statements under independent verification, U1, U3 and U5; two not yet written, U4 and U6; and a condition to be added to a definition. These concern inherited-domain extension, two-cutoff comparison of retained factors, interior backgrounds, corrections near healed regions and interrupted Gaussian factors. The draft says its original U1 form was too strong and uses a corrected form. Group-specific $N \geq 3$ work is separately unperformed. These qualifications belong with any citation of torus uniqueness.

3.4 Navigation through the original mathematics

The raw account is not merely a classical calculation. Its proposed chain contains full densities, large-field histories, source estimates, continuum limits and a witness. These locators lead to actual interfaces:

Conventions and quantifiers. I:11, 401 and 1050 begin Chapters 0, 1 and 2. The action, physical pairing and left-exponential expansion are at I:12756–12760, 15471–15478 and 15643–15648. The last expands at $\exp(i\eta A)U_0$; away from zero it is not the ordinary-potential Hessian without a coordinate-current correction.

Classical response and free currents. I:13681, `4.5:lem-free-current-operators`; I:14496, `4.5:thm-chart-existence`; I:19578, `4.5:thm-operator-bounds`. The domains include graded spaces and independent currents. Our homogeneous physical stationary bounds do not verify all those domains.

Averages and pivots. I:24240, `4.6:lem-averaging-locality`; I:24320–24325 gives corrected composition, I:24349–24369 its flat differential. I:46960 and following gives pivot/constraint linear algebra. II:9032–9075 locates incident-bond Dirichlet geometry, II:9779–9791 its constraint.

Complete representation. I:61097, `4.20:thm-representation`; I:73545, `4.21:rem-global-chain`, specifies four global carriers. I:92401–92620, `4.22:lem-current-localisation`, retains the complex-decoupling current correction.

Sources, per-ball bounds and limits. II:2791, `5.1:thm-correlation-theorem`; II:60348, Chapter 6; II:61205, `6.2:lem-chessboard-estimate`; II:64630, Chapter 7; II:66412, `7.4:thm-infinite-volume`. Its $\mathbb{R}^4$ statement is part of the baseline; it takes sourced estimates as inputs.

Witness, uniqueness, rotations. II:68374 begins Chapter 8; III:5 begins Chapter 9. III:25268, `9.2:thm-carry`, is an incomplete uniqueness-path statement. III:75922 begins Chapter 10; III:76512, `10.1:thm-rotational-identity`. The far-flux assumption reappears at III:110825–110835.

These are navigation points, not exhaustive local proof checks. Shared words such as minimizer, covariance and locality do not identify our results with the source's full-history representation.

3.5 Published results and transfer obligations

Page numbers below are journal pages. Every cited theorem retains its transformation, analytic class, current variables, boundary conditions and coupling hypotheses.

Small fields and extraction. Bałaban, CMP109 (1987), 249–301, https://doi.org/10.1007/BF01215223, Theorem 3, p.264, is the conditional small-field induction for (1.1)–(1.22). Its separate Theorem 2, p.259, announces the tuned $SU(2)$ trajectory. Ward analysis and full Wilson subtraction are in Sections 4–5: (4.9)–(4.15), (4.21)–(4.45), (5.17)–(5.44). Equation (1.19) requires gauge invariance of localized coefficients. A total gauge identity in a different activity representation is not literally that hypothesis.

CMP116 (1988), 1–22, https://doi.org/10.1007/BF01239022, completes the small-field induction. Sections 1–2 localize and integrate the renormalized exponent; (2.41) and pp.21–22 recover the inductive bound and add the Gaussian normalization. This is a substantive closure theorem under small effective couplings, not just a formal loop expansion. Our total-kernel extraction and quotient are alternative arguments with separately stated inputs.

Complete densities. CMP119 (1988), 243–285, https://doi.org/10.1007/BF01217741, defines full pushforward and the additional large-field operation in (0.1)–(0.2); Section 2 specifies its class; Theorem 1 is on p.262. The introductory unnumbered statement is on p.245. Interfaces include (2.18), histories; (2.19)–(2.22), large-field operations; and (2.34)–(2.42), regularity/boundaries. In particular (2.38), p.261, requires local gauge potentials and first derivatives, not merely small plaquettes. General averages satisfying its axioms are allowed, so a composite map is not excluded by name.

CMP122-II (1989), 355–392, https://doi.org/10.1007/BF01238433, Theorem 1, p.355, completes that class under a small common ceiling for all effective couplings. The concluding bounds are pp.390–391, (1.98)–(1.102). Its introduction says the separate perturbative trajectory proof was still unpublished in 1989 and distinguishes later observable analysis. Our search did not locate a later primary proof of that precise theorem; this is a bounded search, not proof of nonexistence. The original CMP122-I formulas were not newly audited in full when its primary PDF endpoint was inaccessible.

Universality and distinct constructions. A familiar coefficient cannot be inserted for an unmatched determinant, cutoff or gauge normalization. One-loop lattice comparison theorems, for example Adams–Lee, https://arxiv.org/abs/0709.0781, have hypotheses on their integrals. They do not directly identify our constrained determinant. Magnen–Rivasseau–Sénoor's construction with an infrared cutoff, CMP155 (1993), 325–383, https://doi.org/10.1007/BF02097397, is a distinct regularized construction. Neither citation is treated as a theorem about our exact macrostep without identification.

3.6 Our additions: objects, results and boundaries

Here addition means a derivation or interface produced in this workspace, not a priority or novelty claim. Gaussian integration, reflection positivity, forest interpolation, cluster expansions, analytic

implicit functions, Ward identities and elliptic estimates remain standard tools. The notebook manifest separates our uses from the cited model-specific results. The documents are research proof drafts with internal audits, not accepted replacements for the literature.

01–05: continuum tools and explicit Gaussian interfaces. Notebook 01 gives a conditional compactness and observable-reconstruction route, factorial-cumulant issues and a stronger positive-energy test. For example, $\langle v, e^{-tH} v\rangle \le q\|v\|^2$, $q < 1$, forces spectral weight at positive energy but neither a gap nor an interacting gauge theory. Notebook 02 develops same-limit diagonal reflection positivity for pure Wilson theory using actual slab gluing and comparison at fixed lattice coupling. Its continuum application still needs common observable limits and the relevant estimates.

Notebooks 03–05 isolate current identities, Gaussian large-field tails, strict background locality under local precision perturbations and simple interaction cases. The forest expansion differentiates Gaussian densities/covariance parameters rather than sharp indicators without justification. Mark localization and strict background dependence are different results. A product domain is stronger than an arbitrary weak source-field ball. These modules do not import every CMP119/CMP122 orbit domain.

06–17: a homogeneous classical and Gaussian base. The flat Maxwell work proves lower bounds, compact right-inverse lifts and upper bounds for the effective quadratic form q_M, uniformly in prior homogeneous refinement M. Alias formulas and pole cancellations give exponential locality and a free refinement limit with $O(M^{-2})$ convergence in the stated weighted norms. These are free-reference limits, not interacting ultraviolet limits.

Subsequent modules identify the source-compatible flat extension and gauge, construct fine inverses and current responses, establish pointwise and equal-exponent regularity, and control nonlinear fixed-reference and moving-reference averages. Ordinary-potential and left-exponential Hessians are related with their current term retained. A small global physical $C^{1,\alpha}$ background admits a local constrained Wilson minimum, gauge-slice coverage and ordinary-constraint transfer. Its classical action obeys $DV = m$; weighted response localizes its Hessian. These statements do not cover arbitrary exterior data, holes, mixed histories or independent free-current domains.

Gaussian roughness is a negative interface that must be retained. In four dimensions, typical fine quantum fluctuations are outside a fixed small physical $C^{1,\alpha}$ ball; the curl trace gives root-mean-square size of order η^{-2}. At a fixed current step, correctly rescaled blocks instead have uniform covariance bounds. Notebook 17 uses this normalization for joint tails. Classical smooth-background estimates cannot simply be applied to a full fine Gaussian sample.

18–22: local integration and strict classical support. Notebook 18 supplies an overlapping-interaction Gaussian theorem with explicit support norms and complex covariance control. Notebook 19 constructs the actual stationary-background pivot chart and reduced precision, retaining constraint curvature and coordinate-current curvature. The independent retained-link window is literal. Haar, ambient straightening Jacobian, exponential-gauge factor and background determinant are separately tracked. Notebook 20 applies these to the bare local step.

Notebook 21 obtains strict support expansions for the classical action through a simultaneous field/multiplier fixed point, using smooth physical cells, mixed insertion bounds and the physical η^4 pairing. It avoids summing M^4 coordinate coefficients and does not assume a uniformly bounded ordinary averaging adjoint where such a bound fails. Notebook 22 transports inherited activities through a stationary map with the explicit multivariate Cauchy cost. Bounded substitution does not automatically contract a norm or preserve a radius through infinitely many steps.

23–28: covariance, re-entry and a common local domain. Notebook 23 treats actual gauge covariance and ordinary representatives. Notebooks 24–26 treat inherited increments, pre-gauge smooth representatives and auxiliary grouping. The physical blocking factor and auxiliary polymer grouping are distinct. The same block labels are not counted twice as newly compressed. Cardinality weights, fiber dimensions, radius loss and thresholds are budgeted explicitly.

Notebook 27 constructs the one-shot constraint $F_R = \mathcal{M}^s_\ell$, $R = \ell^s$, with a selected direct-R contour gauge, actual pivots, measure factors and retained window. Constraint composition does not identify that window with successive primitive windows. Notebook 28 proves a common-domain local integration theorem with joint background/fluctuation supports. An inherited interaction is split into its background value and fluctuation increment; the former has no artificial coupling factor. Equivariant measurable completion is useful for an already defined global invariant density. It does not turn a local analytic germ into the Wilson density outside its domain.

29–31: extraction and finite local induction. Low-degree Ward cancellations are performed on total kernels before localization. Subtracting the full classical reference preserves the nonlinear identity; subtracting only its quadratic jet does not. The total quadratic, cubic and quartic remainders have the low-momentum orders needed by the spatial Taylor argument. Finite-torus versions keep wrapping defects and the actual finite coefficient, rather than imposing an infinite Fourier identity.

The activity quotient identifies normalized totals, tracks vacuum constants, admits controlled periodization/local limits and supports quantum descent. The classical semigroup identifies totals modulo the allowed gauges, not canonical carrier arrays or sequential quantum cutoffs. Finite induction selects equivalent small representatives with strict margins and compensates scalar changes in the vacuum factor. Its initial amplitude theorem was conditional on a weak coupling sequence. Completeness of an analytic interaction space is not construction of a physical probability law.

32–37: the actual determinant and positive local drift. These notebooks retain coarea normalization, fine/target orbit densities, multiplier curvature and actual stationary response. Point-pinned ghosts, globally based quotients and block mean-zero gauges are distinguished. The moving-reference scalar construction yields

$$C - \bar{C} = D_c Z, \qquad \bar{C}D = D_c B, \qquad BB^T = K^{-4} I.$$

This gives an exact gapped covariant determinant pair at fixed K, rather than discarding a singular global-base factor. Soft scalar/vector reductions isolate coarse factors and bounded heat ends. Local trace projections and ordered Duhamel/resolvent products are retained: a local trace does not have unrestricted global cyclicity.

Notebook 37 claims, within the upstream working chain, the actual coefficient estimate

$$\beta(\mathcal{D}_{M,K}) = 22\alpha_g \log K + O(1), \qquad \alpha_g = \frac{C_H}{48\pi^2},$$

uniformly in prior homogeneous depth and every applicable jointly obtainable infinite structural system. Its Maxwell coefficient is twice the normalized affine t^2 coefficient; C_H is the adjoint charge-square sum in the specified color convention. Numerical comparison with another convention requires conversion of action, trace and extraction functional.

The comparison is direct: actual base to Wilson has integrable $O(s^{-1})$ error, and fixed Wilson to continuum has integrable $O(s^{-1/2})$ error. Joint compactness transfers a positive block margin to actual finite coefficients. Real finite-coupling continuity and clipped shooting then produce arbitrarily long finite weak homogeneous local histories with a fixed weak endpoint and bare coupling of order $m^{-1/2}$. This is a substantial local endpoint, not a full-Wilson trajectory theorem, a Schwinger-function limit or a mass gap.

38: exact statements about the full bare measure. Notebook 38 adds two different kinds of result. A specified measurable coarse map admits an exact positive good/remainder decomposition retaining all histories and normalization. Identification of its good term with the analytic germ requires literal containment of the real integration window in the input domain.

Independently, direct compact-group and positivity arguments control the normalized bare Wilson measure. Site-plane reflection positivity and Schatten gluing give joint bad-plaquette bounds. A fixed-box conditional Laplace theorem uses an exact tree quotient, a compact global energy gap and cancellation of Gaussian volume factors to obtain a normalized Ce^{-cT^2} tail. A boundary-only gauge combines these into joint bounds for separated boxes with explicit local minimizer windows. Constants are fixed-box constants, not uniform in growing components.

39: normalized local replacement in the bare Wilson measure. The final extension is `outputs/ym4-normalized-box-replacement/main.tex`. For an arbitrary finite family of vertex-disjoint fixed-size boxes, the replacement preserves the original exterior marginal and raw partition normalization. At a good boundary it conditions the actual conditional law on its minimizer window, dividing by the exact success probability $1 - p_i \geq 1 - a_g$; bad-boundary conditionals are unchanged.

An exact positive failure expansion and coupling control the joint local law, including the exterior. For a selected set of boxes A, the discrepancy is bounded by

$$\mathbb{E}\left[1 - \prod_{i \in A}(1 - p_i)\right] \leq |A| a_g,$$

independently of the total number of replaced boxes. Discrepancy and bad-boundary marks have the stated joint bound $(a_g + \beta_g)^{|S|}$, and bounded local sources/cumulants can be transferred. These are actual bare-measure assertions. They do not concern a nonlinear RG constraint fiber, growing components or an already renormalized density. The final theorem's precise conditions and internal audits remain part of the handoff, rather than being erased by this summary.

3.7 Corrections and superseded routes

Historical notes remain for traceability. Their next-task and candidate language is chronological. Later repairs and counterexamples take precedence. The following changes are particularly important for a reviewer.

1. **No weakened baseline.** The repository did claim $\mathbb{R}^4$ existence. Its definition already excluded a two-point inference of non-Gaussian interaction. Our additions must not be presented against a fabricated weaker starting claim.

2. **The mandatory three-map source obstruction was withdrawn.** An early reading of Raw II Section 5.6 treated three localized response maps as the next unavoidable obstruction. Raw I:73545, `4.21:rem-global-chain`, uses four global carriers instead. The correction is in `work/source_stability_research.md`, opening and Section 9. The conditional local estimates remain; their asserted necessity does not. The two descriptions still need an authoritative use-by-use reconciliation.

3. **Sharp windows were kept distinct.** The retained-link cutoff and additional pivot indicators cannot be conflated. Primary CMP116 p.12 excludes selected pivots from its retained bond set. The nonlinear argument is $\Phi_V(g\xi)$, not generally $g\Phi_V(\xi)$. A bad link implies Euclidean cell norm greater than the same threshold. The separate good-window inclusion with a square-root dimension factor cannot improve the bad-event tail.

4. **Tails are not localized logarithms.** Joint Gaussian bad-block bounds alone do not give connected log activities. Forest/cluster and strict-support arguments are additional work. Grouping arbitrary large overlapping supports into Gaussian blocks can lose degree summability; no routine extension claim substitutes for the actual overlapping-interaction proof.

5. **Norm and radius losses were real.** A fixed cardinality reserve does not pay all Taylor coefficients for free. Transport uses the exact product Cauchy cost, common smaller domains and a geometry budget. Auxiliary grouping/pre-gauge transport address these losses rather than rename them. When auxiliary vertices disappear, connected-hull estimates use Steiner length. Conversion to a terminal metric tree spends a factor of at most two in the exponent.

6. **Positive norm, positive energy and a gap differ.** A nonzero reflected vector can have a zero-energy component. A time correlation ratio forces positive energy but no gap. A constrained Gaussian gap cK^{-2} is not a Yang–Mills Hamiltonian mass gap. Independent random constants and generalized-free examples remain valid continuum-interface cautions.

7. **Pressure and free boundaries did not identify the same symmetric state.** Surface free energy does not determine local states, and translation averaging need not preserve positivity at a fixed mirror. The reflection proof therefore uses actual gluing and comparison with a common-limit obligation, not unrelated states chosen for individual tests.

8. **Gauges and Hessians needed exact corrections.** The source scalar projection is not automatically projection onto $\ker Q_0$; a constrained biharmonic construction was required. Ordinary and left-exponential Hessians differ by current curvature. Constraint elimination adds multiplier curvature. Dropping either changes the determinant.

9. **Pinned and unpinned infrared factors cannot be exchanged.** Point pinning can give a scalar gap of order K^{-4}, not K^{-2}, and its long-time logarithm differs. A globally based scalar factor has a constant-holonomy anomaly with forced vector compensation. The later mean-zero pair is a different exact decomposition, not proof that the earlier factors separately had the desired coefficient.

10. **A false difference-square formula was replaced.** Zero row sums alone do not give the naive component-valued common difference-square identity. Component-skew/odd terms obstruct it. The corrected proof uses ordered Laurent factorization with zero constant and first moments. The amended covariant Hessian and soft-vector/tail modules are operative.

11. **Separate uniform relative vertices were not automatic.** Pure-gauge and hierarchy-interface examples give growing relative vertices while full heat traces remain invariant. Current arguments use covariant densities, localized forms and combined identities rather than the rejected componentwise bound.

12. **Affine probes need a total-kernel interpretation.** A polynomial Cartan probe is not a bounded periodic background. Its use needs localized Taylor kernels and polynomial moments. A convenient local density on an affine field need not equal the total Maxwell moment. Reanchoring, covariance and the limiting order are indispensable.

13. **Infinite kernels are not arbitrary finite kernels.** Large-K magnetic/Bloch arguments use a jointly chosen infinite structural system or explicit periodizations. Finite winding errors and actual coefficient transfer are separate. Finite determinant algebra alone does not justify thermodynamic derivative interchange.

14. **Full-measure shortcuts fail.** Small plaquettes on a simply connected box give a tree-gauge link bound proportional to a filling area. Alternating small curvature rules out an automatic mesh-uniform physical $C^{1,\alpha}$ chart. Flat torus holonomies obstruct an arbitrarily small global potential. Raw link marginals are Haar by gauge invariance, so raw near-identity link probability is the wrong event.

3.8 Remaining existence interfaces and audit priorities

The local positive drift changes the next research task; it does not erase the remaining implications.

Full densities and normalized large-field replacement. One needs a full match to the CMP119/CMP122 class or a direct replacement theorem for the chosen composite map. It must handle relevant bad components, exterior data, holes, retained terms, interrupted operations and ages, with numerator/denominator cancellation and summable bounds. The weak couplings of our local recursion are not automatically the coefficients of that full recursion. The averaging axioms may permit the map, but all used variational/domain hypotheses still require verification.

The new normalized fixed-box replacement is a useful concrete piece of this problem. It does not identify fixed-boundary conditionals with conditionals on nonlinear block averages, or establish iteration on post-RG densities. An exact all-history identity is not a uniform norm bound on those histories.

Observables and a common continuum limit. An external observable source changes full densities, counterterms and large-field operations. Its cutoff/volume uniform control is not implied by holomorphy in a small background connection. Analytic activity compactness is not tightness of a quantum observable law. A common hierarchy must retain consistency, positivity, symmetries and the growth required by the intended reconstruction. Coincidence extensions and the observable algebra must be specified.

Identification and infrared properties. A scalar action-density subsector with a running two-point witness is not alone identification of an interacting Yang–Mills theory. Clustering, vacuum uniqueness and a mass gap are separate infrared questions. The source draft expressly leaves them open; our local ultraviolet work does not close them. General color formulas and bare $SU(N)$ probability lemmas do not extend every upstream $SU(2)$ input to all compact simple groups.

Efficient audit order. First freeze the target theory and authoritative companion revision. Verify normalizations, finite measure identities and domains before long estimates. For our series, inspect the classical base, strict support norms, common-domain quantum theorem, finite extraction/ induction, determinant reduction/ comparison, then finite drift transfer. Audit the full-measure interface independently; the length of the local chain does not prove it. Historical audit files guide checks but are not independent certificates.

3.9 Companion files and citation conventions

This report has a plain-text counterpart, `work/handoff_repository_baseline.txt`, and a machine-readable source map, `work/handoff_repository_source_map.json`. The current detailed obligation ledger is `work/final_push_existence_dependency_audit.txt`. Primary source maps include `work/renormalized_induction_primary_sources.md`, `work/full_measure_large_field_interface_research.txt`, and `work/mean_zero_one_loop_source_interface.txt`. The source-route

correction is in `work/source_stability_research.md`. The notebook manifest identifies authoritative assembled files and the final extension's audits.

The repository and published articles should be cited for their own results and definitions. Our claims should be cited as research drafts with hypotheses. This distinction remains necessary even where a calculation seems elementary or several internal readers agree. The handoff supplies a reproducible theorem-level map, new explicit derivations and remaining interfaces. It does not declare the existence problem solved.

4 The local construction added in this project

The main route is

$$\begin{gathered}\text{flat coercivity and classical minimizers}\\ \longrightarrow \text{actual local quantum integral}\\ \longrightarrow \text{Ward extraction and finite induction}\\ \longrightarrow \text{actual positive determinant drift}\\ \longrightarrow \text{arbitrarily long finite weak local histories.}\end{gathered}$$

The catalogue supplies every intermediate notebook, rather than replacing this chain by an appeal to perturbation theory.

4.1 Flat and nonlinear classical analysis

Notebooks 06–17 develop flat coercivity, constrained lifts, weighted inverse kernels, refinement limits, pointwise regularity and nonlinear response estimates for the homogeneous classical minimization problem. The construction tracks physical pairings, contour constraints, gauge directions, and the relation between a prescribed coarse field and its fine minimizer.

Their endpoint is a family of local minimized classical actions S_M, analytic charts, and localized response kernels near the trivial orbit. The support estimates matter because quantum integration must preserve spatial summability, not merely analyticity on a finite-dimensional ball.

These results concern regular homogeneous backgrounds. They do not give a minimizer theorem on every determining set with holes, arbitrary exterior boundary fields, or independent complex currents. Their estimates cannot be used on those domains by renaming variables.

4.2 The actual local quantum integral

Notebooks 18–28 formulate and control one fixed composite ordinary coarse map $F_R = \mathcal{M}_\ell^s$, its chosen direct-R contour gauge, and its retained-coordinate small-field sector. The nonlinear constraint chart $\Phi_V(z)$ keeps the retained coordinates unchanged and solves the pivot variables. Its integration field is

$$U = e^{i\Phi_V(g\xi)}V,$$

not $e^{ig\Phi_V(\xi)}V$. This distinction affects higher-order interactions and the exact Jacobian.

The pullback includes Haar densities, the square ambient straightening, the pivot constraint derivative, the contour gauge penalty, the actual reduced Hessian, and the independent retained link/color-ball window. A rectangular determinant of the retained embedding cannot replace the constraint Jacobian.

The compact gauge insertion uses the exact normalizer

$$c(g) = \int_G \exp[-(1 - \operatorname{Re}\tau h)/g^2]\, dh.$$

For a literal represented input density the local output has the following factorization, with notation abbreviated but the normalization displayed:

$$Z_{\text{in}}\, c(g)^{-n_s} h_0^{n_e-n_c} D_R(V)\, g^{n_z} (2\pi)^{n_z/2} \det A_M(V)^{-1/2} \\ \times e^{-S_M(V)/g^2 + E_{\text{raw}}(V)} \mathbb{E}_{A_M(V)^{-1}}[\chi_{\text{window}} \chi_{\text{extra}} e^{I(V,g,\xi)}].$$

Here $n_z = (\dim \mathfrak{g})(n_e - n_c)$, D_R is the actual pivot factor, h_0 is the Haar density at the origin in the chosen Lie coordinates, and I contains all higher action, Haar, straightening and inherited fluctuation terms. Individual notebooks define every factor without shorthand.

The common-domain argument prevents uncontrolled shrinkage of the analytic field ball at every step. It uses the actual pre-gauge representative and transports the *entire* inherited insertion, including its background value. It does not merely change a fluctuation variable and forget its cutoff. Strict connected support families and overlapping-carrier expansions provide the spatial norms.

Composition of primitive constraints identifies the ordinary coarse map. It does not identify one composite retained window with an intersection of sequential primitive windows. Also a holomorphic continuation of a local integral is not a representation of the full positive measure outside its real integration domain.

4.3 Ward extraction and finite local induction

Notebooks 29–31 use Ward identities of the total functional to isolate its marginal Maxwell contribution. They do not assume that each arbitrarily selected activity is separately gauge invariant. The remaining interactions are relocalized and controlled in a suitable representative of the exact finite total.

That representative need not be the original canonical cluster array. Equivalence of the total and its compensating vacuum constant are tracked: changing E to $E + c$ requires changing Z to Ze^{-c}. Dropping that factor would destroy a later full-measure normalization comparison.

The finite induction bounds the local remainder for permitted weak coupling sequences and sufficiently large compatible periods. It does not itself force the couplings to remain weak. Positive drift and finite shooting supply that part.

4.4 The actual logarithmic drift

Notebooks 32–37 compute the leading coefficient for the actual constrained determinant. An exact soft reduction expresses it as a fine scalar/vector pair and coarse factors. The coarse Maxwell contributions and both heat-integral ends are bounded uniformly in preceding depth.

On the remaining band $1 < s < K^2$, source-localized resolvents remove soft penalties with an integrable error. Nonlinear Ward jets through field degree four compare the actual effective-action vector base with the Wilson base; left-relative chart-current terms remain present. A direct Wilson-to-continuum comparison controls the remaining error and fixes signs and color multiplicities.

With $\tau(H^2) = 1$ and the full adjoint charge trace,

$$C_H = \sum_{q \in \text{spec}(\text{ad}_H)} q^2, \qquad \alpha_g = \frac{C_H}{48\pi^2},$$

the working theorem is

$$\beta(\mathcal{D}_{M,K}) = 22\alpha_g \log K + O(1).$$

The error is uniform in old depth and every applicable joint infinite structural system, under the classical and local quantum inputs explicitly used in the series. For $\tau = \text{Tr}\,/N$, the color calculation gives $C_H = 2N^2$; this does not establish earlier construction inputs for every $SU(N)$.

Here $\beta = 2[t^2]$ on the normalized affine probe. The signed contributions of the mean-zero scalar and vector factors to $\mathcal{D}$ have coefficients $2\alpha_g$ and $20\alpha_g$. An older point-pinned ghost decomposition distributes logarithms differently; its scalar coefficient cannot be combined with the new mean-zero vector factor.

4.5 Finite weak histories and the recursion actually tuned

Finite moment transfer converts the positive margin in every joint structural limit to a uniform finite block margin. The geometric ratios are fixed first; a block of q steps then dominates the bounded comparison error; periods and the weak ceiling are chosen so the actual errors do not consume that margin.

A continuous clipped finite recursion is shot to a fixed weak endpoint Γ_*, where $\Gamma = g^{-2}$. Positive future blocks show clipping did not occur on the selected trajectory. With m blocks,

$$(\Gamma_* + mqD_{\mathrm{abs}})^{-1/2} \le g_{\mathrm{bare}} \le (\Gamma_* + m\delta/4)^{-1/2}.$$

This supplies arbitrarily long finite *homogeneous local* weak histories. It does not claim uniqueness, consistency between horizons, or that a full large-field density has the same extracted recursion. That identification remains open.

5 The full-measure additions

Notebooks 38–39 add exact finite positive-measure results. Their bare Wilson probability theorems are independent of the earlier determinant work. The local-germ embedding uses the already established coarse chart and Ward identities.

5.1 Exact positive history operators

Fix a measurable equivariant completion of the actual local coarse map, using its original mean on its good set and its original reference string outside. This specifies a global coarse observable of the original fine measure; it does not invent a global analytic effective action.

The compact based-gauge insertion gives a probability kernel preserving the coarse field. An actual retained sheet yields a local density with

$$F_*\mu \ge 1_D \ell_\mu \, m_C.$$

For a nonnegative cutoff θ supported in D, let $m_\theta(W) = \int \theta(W^v)\, dv$. Define

$$\mathcal{A}_g\mu(dW) = \frac{\int \theta(W^v)\ell_\mu(W^v)\, dv}{m_\theta(W)}\, m_C(dW),$$

with value zero when $m_\theta = 0$. The invariant absolutely continuous density of the true pushforward bounds every numerator orbit value. Hence

$$0 \le \mathcal{A}_g\mu \le F_*\mu, \qquad \mathcal{R}_g = F_* - \mathcal{A}_g \ge 0.$$

These linear invariant operators retain singular parts in the remainder. Their finite products account for every history, raw mass and observable numerator exactly.

Local-germ equality uses a finite orbit-overlap lemma and the literal invariant-ball cutoff Ward identity. Its radius may shrink with graph diameter. Every reconstructed real field must also lie where the input measure actually has the claimed density. Holomorphic continuation alone supplies neither condition.

5.2 Actual Wilson rarity and conditional normalization

For site-plane cells of fixed side $b \geq 3$ and torus periods $n_i = 2m_i b$, exact Haar slab gluing and even Schatten Hölder give a chessboard inequality. A lower bound on the *full* partition function yields

$$\mu_g\left(\bigcap_{j\in S} B_j(\delta)\right) \leq z_b(g,\delta)^{|S|},$$
$$z_b = \min\{1, e^{-\delta^2/g^2 + b^d(d\dim G\log(1/g)+C_{G,d})}\}.$$

Each B_j forces an interior plaquette energy $1 - \operatorname{Re}\tau U_p \geq \delta^2$. Finite coloring gives prescribed-plaquette bounds with exponent loss $Q_b = b^d\binom{d}{2}$. This is a normalized interacting Wilson estimate, uniform in surrounding volume.

For a fixed box $[0,R]^4$ with small prescribed boundary edges, a forest rooted at every boundary vertex removes the interior gauge redundancy exactly. At identity boundary its representative has a unique zero-action field and a positive Hessian. Compactness and the implicit-function theorem give a unique global conditional minimum for nearby data. Numerator and denominator have the same local factor g^n; the outer compact region has a fixed energy gap. Thus

$$\mu_{V,g}(W^c_{V,g,T}) \leq Ce^{-cT^2}, \qquad 1 \leq T \leq r/g.$$

The window uses an adjoint-invariant edge-max norm about that conditional minimizer.

The boundary complex is S^3. Boundary tree loops have finite boundary-plaquette fillings, so small boundary curvature supplies a boundary-only small gauge. Conditioning on the exterior of vertex-disjoint boxes makes their interiors independent. Combining the estimates gives

$$\mu_g\left(\bigcap_{i=1}^m D_i\right) \leq \left[Ce^{-cT^2} + 24R^3 z_b(g,\varepsilon_R)^{1/Q_b}\right]^m.$$

Here D_i is a bad boundary or a failed local gauge/minimizer window. It is an event in the complete bare measure. It is not yet the nonlinear averaged constraint window of the iterated map.

5.3 The final normalized replacement

Notebook 39 conditions on the common exterior ω. The interior laws factor as $\prod_i Q_i^\omega$. On a good boundary put

$$p_i(\omega) = Q_i^\omega(W_i^c) \leq a_g, \qquad P_i^\omega = Q_i^\omega(\,\cdot \mid W_i).$$

On a bad boundary take $P_i^\omega = Q_i^\omega$ and $p_i = 0$. The operation does not remove bad boundaries.

For the original exterior marginal ρ, set

$$\nu(d\omega, dx) = \rho(d\omega)\prod_i P_i^\omega(dx_i).$$

This preserves probability and, in the unnormalized version, the original raw partition mass. Its Radon–Nikodym factor is a product of good-window indicators divided by their actual probabilities $1 - p_i(\omega) \geq 1 - a_g$.

An explicit coupling changes an interior only on a conditionally independent failure of probability p_i. For the exterior jointly with a selected set A, total variation, with convention $\sup_E |\mu(E) - \nu(E)|$, is exactly

$$\mathbb{E}_\rho\left[1 - \prod_{i\in A}(1-p_i)\right] \leq |A| a_g.$$

The estimate is independent of the number of other boxes. The full theorem gives sparse discrepancy bounds, retains correlated bad boundaries, and transfers bounded local sources with the same normalized laws.

This completes a finite localized replacement theorem. It does not cover arbitrary growing components, repeated scales, or nonlinear coarse-average fibers. Preserving normalization does not establish that the homogeneous coupling recursion remains unchanged.

6 The remaining theorem program

6.1 H2 Complete-density transfer for the chosen map

The first major unresolved theorem is propagation of a complete density for the same full Wilson measure and coupling recursion as the local construction. There are two routes.

An import route verifies the entire induction class of Bałaban's complete-density theorem for the chosen coarse map and operations. That theorem is conditional on its own effective couplings staying weak. Matching their expected size or one-loop sign is not a match of the recursion. The primary-source map appears in the baseline assessment and Notebook 38.

A direct route builds a spatial determining-set decomposition with collars, a regular constrained background, rough fluctuations and exceptional regions. It proves exact good-component integration with the actual window and Haar, gauge, pivot and vacuum factors. It then bounds normalized replacement on every allowed complementary component, with constants summable in geometry and scale age. Notebook 39 supplies a restricted fixed-box example.

The proof should state admissible exterior fields, transition or holonomy data, constraints, conditional measures, analytic domains and the component norm. Its denominator lower bound must be compatible with the numerator's boundary energy. An unrelated Gaussian normalization cannot be inserted after conditioning.

Several shortcuts are excluded. Unfixed raw link marginals are Haar. Flat periodic fields can have nontrivial holonomy. Even plaquettes of size $O(\eta^2)$ can have alternating physical curvature whose adjacent norms prohibit every uniformly bounded $C^{1,\alpha}$ gauge. Small curvature alone is not a smooth-background construction.

6.2 H3 Observable and source estimates

After complete-density control, external source estimates are needed for a specified gauge-invariant observable algebra. Analyticity in a connection background is not analyticity in an external composite-observable source. The latter changes the density, local counterterms, translation properties and possibly its large-field estimates.

A useful first target is renormalized action density on reflected separated test supports. Its source neighborhood must be uniform in mesh and volume in the exact cells used by reflection. Mixed derivatives must obey all-order distributional bounds, for example factorial bounds in stated test-function seminorms. Fixed-point-number bounds with no controlled growth in that number do not supply field reconstruction estimates.

Numerator and denominator must belong to the same positive regulator construction. Notebook 39's bounded-source theorem does not control renormalized fields whose supremum norms grow with refinement.

6.3 H4 A common continuum hierarchy

Once actual model bounds are available, extract one diagonal subsequence for a countable dense test algebra and extend by uniform continuity. Products, mixed insertions, subtraction conventions and

required contact extensions must be compatible on that same subsequence. Compactness of analytic interactions on a background ball is a different problem from tightness or distributional compactness of physical correlations.

Separated correlations may not determine every distribution needed for local fields. A route avoiding particular coincidence extensions should state why; otherwise those extensions and counterterms must be constructed compatibly with positivity and covariance.

6.4 H5 Positivity, rotations and reconstruction

Reflection positivity must descend from the same positive finite measures and limiting observable algebra. Estimates for a substitute measure or unrelated subsequence cannot simply be combined with it.

Hypercubic symmetry is not $O(4)$ covariance. The reflection-based restoration mechanism has conditional geometric and source-cell reductions in the early notebooks. Its uniform model estimates still need to hold for all transformed supports, cell positions and all-order constants used by that argument.

Osterwalder–Schrader reconstruction requires the chosen hierarchy's precise regularity and positivity hypotheses and an appropriate vacuum sector. Unique vacuum or clustering needs its own estimate. If existence can use a suitable sector or decomposition instead, that choice should be explicit.

6.5 H6 Nontrivial Yang–Mills identification

A positive beta coefficient does not establish nonzero positive-energy states in the reconstructed theory. A quantitative lower witness must survive the same limit. In a reconstructed theory, a strict inequality

$$\|\psi\|^2 - \langle \psi, e^{-tH}\psi \rangle > 0, \qquad t > 0,$$

would certify positive-energy content. The task is to prove such a witness from the cutoff theory.

A non-Gaussian scalar composite is not by itself an interaction witness: a quadratic composite of a free Gaussian field is already non-Gaussian. The algebra, tensor laws and short-distance normalizations must be sufficient to identify the limit as Yang–Mills. A sufficient identification criterion should be stated and proved; no speculative criterion is promoted to a theorem in this dossier.

6.6 H7 Physical mass gap and group scope

The gap in an auxiliary constrained Gaussian, typically at the blocking scale, is not a positive gap in the interacting Hamiltonian above the vacuum. A physical infrared estimate must be uniform in the same continuum sequence. An $SU(2)$ construction does not settle the all-compact-simple-group target. These are separate from finishing the ultraviolet program.

7 How a successor can use the bundle

7.1 A recommended first pass

Read the start document, the baseline assessment and catalogue entries 31–39. Then inspect Notebook 37's actual coefficient theorem and trace its upstream inputs, or independently begin with Notebooks 38–39's bare Wilson arguments. These routes need different expertise and can be reviewed in parallel.

For the drift, prioritize relative Hessian/current terms, nonlinear Ward jets, the determinant decomposition, infrared and source-localized traces, and the transfer from joint structural limits to finite drift. For complete-density transfer, prioritize domains, coarse-map completion, retained-window equality, exterior normalization and coupling extraction.

7.2 A concrete next research deliverable

The next useful deliverable is a theorem for a collared component with the actual averaged constraint. It should specify:

1. the component, exterior data, gauge/transition variables and compact conditional integral;
2. the constrained background and retained map with its full Jacobian;
3. an exact normalized replacement or good/remainder identity preserving the fine measure when components are summed;
4. a bound in the complete-density induction's actual norm;
5. compatibility with neighboring regions, scale changes, sources and coupling extraction.

Verifying the published induction class may be more efficient than rebuilding its geometry. Choose by an exact hypothesis comparison. A direct construction changing the coarse map must redo its local identification and drift transfer.

7.3 Review and reproducibility

Internal audits and successful compilation are recorded, not called external certification. A successor should look for hidden domain enlargement, dependence on a supposedly uniform parameter, an unrelated conditional denominator, or identities combined from different limits. These are specific failure modes encountered and repaired during this project.

Portable sources contain the assembled notebooks and historical fragments. Reproduction instructions give compile commands; the manifest records delivered bytes. Third-party articles are linked rather than republished. The repository can be fetched at its pinned commit and checked against the recorded tree.

The outcome is a detailed foundation and continuation program. The ambition remains existence; the bundle makes the difference between that ambition and the established arguments inspectable.